

ARTIFICIAL INTELLIGENCE ASSESSMENT – 3 REPORT

Course Code: CSA1717 – Artificial Intelligence

Submitted By

Name: Monika M

Register Number: 192521435

Department: Information Technologyro

Acknowledgement

I express my sincere gratitude to our faculty member for providing the opportunity to complete this Artificial Intelligence laboratory exercise. I am thankful for the valuable guidance, encouragement, and support received throughout this work. This experiment helped me understand the implementation of AI search algorithms and game-playing algorithms using Python. I also thank my institution for providing the necessary resources and facilities to complete this laboratory exercise successfully.

Introduction

Artificial Intelligence (AI) enables computers to solve problems, make decisions, and find optimal solutions similar to human intelligence. AI algorithms are widely used in pathfinding, robotics, navigation systems, and game-playing applications. In this laboratory exercise, two important AI algorithms are implemented: the A* Search Algorithm and the Minimax Algorithm with Alpha-Beta Pruning.

The A* Search Algorithm is an informed search technique that finds the shortest path between a start node and a goal node using both the actual path cost and heuristic values. The Minimax Algorithm with Alpha-Beta Pruning is used in game-playing applications to determine the best possible move while reducing unnecessary computations through pruning. These algorithms demonstrate efficient problem-solving and decision-making techniques in Artificial Intelligence.

Experiment 1: A* Search Algorithm

Problem Statement

To implement the A* Search Algorithm for finding the optimal path from the start node (A) to the goal node (G) using heuristic values and path costs. The algorithm should calculate the shortest path and its total cost.

Solution

The graph was represented using an adjacency list with edge costs, and heuristic values were assigned to each node. The A* Search Algorithm calculates:

- $g(n)$: Actual cost from the start node to the current node.
- $h(n)$: Estimated cost from the current node to the goal node.
- $f(n)$: Total estimated cost, calculated as $f(n) = g(n) + h(n)$.

The node with the lowest $f(n)$ value is selected at each iteration until the goal node is reached. Python was used to implement the algorithm using a priority queue (heap).

Result

The program successfully identified the optimal path from node A to node G.

Optimal Path: $A \rightarrow B \rightarrow E \rightarrow G$

Total Path Cost: 6

The algorithm correctly selected the shortest path based on the evaluation function and heuristic values.

Experiment 2: Minimax Algorithm with Alpha-Beta Pruning

Problem Statement

To implement the Minimax Algorithm with Alpha-Beta Pruning for a game tree and determine the best move for the MAX player while reducing unnecessary node evaluations through pruning.

Solution

The game tree consists of one MAX node and two MIN nodes. The algorithm evaluates the leaf nodes from left to right.

The implementation follows these steps:

- The MIN nodes choose the minimum value among their children.

- The MAX node selects the maximum value returned by the MIN nodes.
- Alpha (α) stores the best value found for MAX.
- Beta (β) stores the best value found for MIN.
- If $\beta \leq \alpha$, the remaining branches are pruned since they cannot influence the final decision.

The program was implemented in Python and accepts different leaf values as input.

Result

The program successfully evaluated the game tree and selected the optimal move.

For the given input:

- Left MIN Values: 3, 5, 6
- Right MIN Values: 9, 1, 2

The obtained results were:

- Best Move for MAX: Left Subtree
- Final Minimax Value: 3
- Pruned Node: 2

The implementation correctly demonstrated Alpha-Beta Pruning by eliminating unnecessary node evaluation and reducing computation time.

Conclusion

The objectives of both experiments were successfully achieved. The A* Search Algorithm efficiently found the shortest path using heuristic information, while the Minimax Algorithm with Alpha-Beta Pruning effectively selected the optimal move and reduced the number of nodes evaluated. These experiments provided a practical understanding of search algorithms and decision-making techniques used in Artificial Intelligence applications such as navigation systems, robotics, and game-playing AI.